

# DRL-Driven Optimal User Association and Load Balancing in Hybrid RF/LiFi-Based IoT Systems

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**Abstract**—The proliferation of Internet of Things (IoT) devices has raised considerable difficulties in the identification of users, optimization of power, and load balancing in hybrid RF/LiFi networks. As interconnection among devices increases, ensuring optimal performance while managing network resources efficiently becomes quite complex. This complexity arises due to accommodating the possibly diverse users' needs, fluctuating channel conditions, and varying interference levels, all necessitating sophisticated management solutions to provide seamless connectivity and dependable communication. To tackle these issues, a deep joint hybrid system (DJHS) technique is presented, which employs proximal policy optimization (PPO), a cutting-edge deep reinforcement learning (DRL) algorithm. DJHS aims to effectively handle the intricate problems surrounding user association and load balancing while optimizing power usage in dynamic contexts. DJHS continuously updates its approach based on real-time network data through adaptive learning methods, allowing it to make intelligent decisions that improve overall system performance regarding data throughput and power optimization. Simulation results demonstrate that DJHS outperforms existing approaches such as soft actor-critic (SAC), advantage actor-critic (A2C), twin-delayed deep deterministic (TD3), and trust region policy optimization (TRPO) regarding crucial metrics, including data rate and power transmission. Notably, DJHS's ability to adjust to variations in signal-to-interference-plus-noise ratio (SINR) allows for enhanced resource allocation and network stability. This flexibility ensures that users receive optimal service even in changing conditions, enhancing the overall user experience.

**Index Terms**—Light fidelity (LiFi), hybrid RF/LiFi, Internet of Things (IoT), deep reinforcement learning (DRL), deep joint hybrid system (DJHS), user association, transmission power allocation, load balancing, throughput.

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## I. INTRODUCTION

RADIO frequency (RF) communication has been the backbone of wireless connectivity for decades, offering benefits such as vast coverage and portability [1]. However, it is restricted by a few shortcomings such as congestion and interference affecting the performance in crowded Internet of Things (IoT) environments [2]. As the number of devices grows, it leads to a bottleneck in bandwidth that causes an increase in latency and subsequently lowers data rates as well as performance. Also, the noise introduced by other RF devices can significantly hamper signal quality [3]. Aside from congestion, the RF spectrum is a precious medium that has been auctioned and assigned by powerful regulatory methods [4]. With the advent of IoT, the RF bandwidth has seen a significant. As a result, leveraging devices is now possible only to a limited extent in demand [5]. Another significant aspect of RF communication. Many RF communication protocols, particularly those designed for high-throughput applications, can be less power-efficient compared to low-power protocols like Bluetooth Low Energy, depending on their specific use case and configuration. A continuous high energy demand can shorten the service life of devices and lead to higher operating costs [6].

To overcome the shortcomings of RF communication, light fidelity (LiFi), a form of visible light communication (VLC), is an innovative wireless technology that uses light to transmit data at high speeds [7]. While RF has data rate limitations, LiFi can reach higher bandwidth and thus is an excellent solution for high-speed connectivity applications like augmented reality or high-definition video streaming. In addition, as LiFi works in the visible light spectrum, it does not interfere with RF devices, providing a high level of communication reliability even when coexisting with RF devices. LiFi uses existing lighting infrastructure, which removes the necessity for extra equipment that consumes energy, often beneficial to IoT apparatus needing a bulky battery source [8]. Furthermore, the joint use of RF and LiFi technologies in an integrated communication system takes advantage of the benefits offered by each of them [9], [10].

The user association process allows the devices in IoT to connect to a network or an access point receive data from it [11]. In a hybrid RF/LiFi system, the association's main functions are to manage efficient communication and also maintain the quality of service (QoS) [12]. Given the dynamic nature of IoT environments (QoS), a highly sophisticated mechanism is needed for maintaining an association that can

also dynamically adapt [5]. However, the devices moving in and out of RF network coverage or transitioning between access points may experience delays due to protocol overhead when connecting to the access gateway.

Another critical factor is that environmental changes, such as obstacles and light sources or user density, affect both RF and LiFi network performances. Additionally, diverse IoT applications have different QoS requirements and face challenges such as managing traffic density and resource diversity. For instance, Mohajer et al. [13] propose FlexSlice for mobile edge networks, using graph attention and deep reinforcement learning to improve traffic prediction and offloading decisions. Hence, the process of establishing a relationship is more complex. To overcome these challenges, various methods for association have been proposed, including signal strength-based association, load-aware association, and context-aware handover mechanisms. In hybrid RF/LiFi systems, association and load balancing techniques are critically essential to ensure the maximum accuracy of IoT networks. As IoT grows ever wider, advancements in this field can result in significant benefits, including improved network performance, scalability, greater energy efficiency, and a more seamless user experience [14], [15].

#### A. Motivation

The increasing deployment and adoption of IoT devices result in novel issues concerning stable and efficient communication in high-traffic environments that undergo high variation. Conventional communication methods suffer from congestion, interference, and limited bandwidth. On the other hand, LiFi technology offers high-speed data transmission but is limited by its short range and is dependent on the environment. Moreover, managing user association and load balancing in the hybrid IoT environment is a performance bottleneck. However, as discussed above, a novel and intelligent approach is still required to resolve the issues mentioned above. Therefore, we design a deep joint hybrid system (DJHS) strategy based on the deep reinforcement learning (DRL) approach using the proximal policy optimization (PPO) algorithm to enhance network performance.

#### B. Contribution

The key contributions of this study are as follows.

- 1) This study presents a novel user association and load-balancing algorithm to balance resources between RF and LiFi optimally, that is unaffected by the dynamics of IoT networks. The proposed approach reduces network congestion by efficiently integrating RF and LiFi communication, maximizing overall network capacity.
- 2) We develop a deep reinforcement learning-based DJHS framework using PPO to adaptively associate IoT devices with optimal in an integrated RF/LiFi environment. This mechanism dynamically adapts to real-time user mobility, changes in signal conditions, and application-specific requirements to optimize connectivity while improving QoS.

- 3) The proposed method enables real-time decision-making to constantly inspect network conditions and adjust user associations and load distributions.
- 4) The proposed framework improves energy-utilization communication. It includes experimenting, assessing access points, and tuning load distribution to enhance the throughput of devices and save operational costs by reducing energy consumption.
- 5) The extensive simulation and results show the efficiency of the proposed method and outperform other existing deep learning algorithms in hybrid scenarios.

In summary, the proposed (DJHS) method offers distinct advancements in user association and load balancing for hybrid RF/LiFi IoT systems by adaptively learning and optimizing in real-time using the customized PPO algorithm. This stability enables DJHS to handle large-scale IoT networks' high-dimensional state and action spaces, addressing scalability issues and limitations discussed in Section II and Table I. However, the proposed method improves overall system performance in data throughput, power optimization, and user satisfaction. The remaining sections of this paper are structured as follows: Section II discusses the related works. Section III discusses the system model. Section IV introduces the proposed DJHS framework. Section V discusses performance evaluation and presents the simulation results. Finally, we conclude this work in Section VI and set the ground for future work.

## II. RELATED WORKS

This section explores the literature review of the recent advancements in user association and load-balancing strategies for hybrid RF/LiFi systems.

#### A. User Association in Hybrid RF/LiFi Systems

The user association process associates users with the access points (APs) to optimize network throughput and performance. As RF and LiFi technologies differ in many aspects, user association in hybrid RF/LiFi systems is crucial. Machine learning methods have recently been used to enhance user association. The authors in [16] proposed a machine learning-based user association strategy in the IoT environment, thereby improving the quality of connection and low latency significantly. Through explicit real-time user-to-AP quantification and design, the potential of reinforcement learning-based user association is leveraged to minimize the objective in IoT systems, enhancing network efficiency.

In [17], another framework, which couples user mobility prediction and reinforcement learning, was proposed for user association. Here, seeking knowledge of user movement assists the proposed technique in reducing handover times and enhancing general network efficiency. This approach highlights the importance of including proactive models for facilitating user engagement in dynamic IoT environments. A more powerful tool for optimizing user engagement effectiveness is reinforcement learning (RL). RL has been employed in [18] to optimize user associations according to the instantaneous state of the network. The authors showed that the

TABLE I  
SUMMARY OF THE RELATED WORKS

| Reference         | Method                                                                   | Key Features                                                                                                                                                                                                 | Limitations                                                                                                                                                                 |
|-------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [15]              | Machine Learning-based User Association                                  | <ul style="list-style-type: none"> <li>Significant improvement in connection quality</li> <li>Reduced latency</li> </ul>                                                                                     | <ul style="list-style-type: none"> <li>Sensitive to environmental changes</li> <li>Requires extensive training data</li> </ul>                                              |
| [16]              | Mobility Prediction and Reinforcement Learning                           | <ul style="list-style-type: none"> <li>Reduces handover times</li> <li>Enhances network efficiency</li> </ul>                                                                                                | <ul style="list-style-type: none"> <li>Complexity in accurately predicting user mobility</li> </ul>                                                                         |
| [17]              | Reinforcement Learning                                                   | <ul style="list-style-type: none"> <li>Outperforms conventional techniques</li> <li>Improves user satisfaction</li> <li>Increases throughput</li> </ul>                                                      | <ul style="list-style-type: none"> <li>Requires constant state monitoring</li> <li>May struggle with dynamic environments</li> </ul>                                        |
| [18]              | Deep Q-Learning                                                          | <ul style="list-style-type: none"> <li>Maximizes connected users</li> <li>Balances resources</li> <li>Increases data rates</li> </ul>                                                                        | <ul style="list-style-type: none"> <li>High computational cost</li> <li>Performance may degrade in highly variable conditions</li> </ul>                                    |
| [19]              | Traditional Load Balancing Techniques                                    | <ul style="list-style-type: none"> <li>Improves system efficiency</li> <li>Increases user satisfaction</li> </ul>                                                                                            | <ul style="list-style-type: none"> <li>Often static</li> <li>Not adaptable to dynamic network conditions</li> </ul>                                                         |
| [20]              | Live Traffic Monitoring and Responsive Allocation based on Deep learning | <ul style="list-style-type: none"> <li>Increases throughput</li> <li>Enhances customer satisfaction</li> </ul>                                                                                               | <ul style="list-style-type: none"> <li>May not handle rapidly changing user needs effectively</li> </ul>                                                                    |
| [21-23]           | Deep Reinforcement Learning                                              | <ul style="list-style-type: none"> <li>Increased stability</li> <li>Congestion reduction</li> </ul>                                                                                                          | <ul style="list-style-type: none"> <li>High processing requirements</li> <li>Sensitive to hyperparameter tuning</li> </ul>                                                  |
| [24-29]           | Hybrid Load Balancing Scheme                                             | <ul style="list-style-type: none"> <li>Enhances resource allocation efficiency</li> <li>Increases data rates</li> <li>Increases user capacity</li> </ul>                                                     | <ul style="list-style-type: none"> <li>Complexity in integrating RF and LiFi features</li> <li>Needs effective resource management</li> </ul>                               |
| [30-31]           | Traditional SAC, TD3, and A2C Algorithms                                 | <ul style="list-style-type: none"> <li>Potential for adaptive learning in resource allocation</li> </ul>                                                                                                     | <ul style="list-style-type: none"> <li>High processing demands</li> <li>Struggles with scalability and communication costs</li> </ul>                                       |
| [Proposed (DJHS)] | Adaptive Learning, DRL using PPO                                         | <ul style="list-style-type: none"> <li>Optimized Real-time data,</li> <li>Optimizes power usage,</li> <li>reduced operational costs.</li> <li>Scalable and Robustness in a hybrid IoT environment</li> </ul> | <ul style="list-style-type: none"> <li>May demand significant computational power for training</li> <li>May require extensive resources depending on environment</li> </ul> |

proposed approach outperformed conventional techniques in terms of user satisfaction and network throughput. In addition, the author proposed a new user association scheme based on deep Q-learning to maximize the number of connected users in hybrid RF/LiFi networks [19]. The results confirmed that the proposed method accommodates users while maintaining a balance with network resources, consequently increasing data rates while reducing latency.

### B. Challenges in User Association

Traditional methods advance user association performance, but many issues still need improvement. The main challenge is the difference in channel conditions, which results in variable signal quality and user experience quality. In addition, the

handover process, which refers to moving a user from one AP to another, incurs delays rather than providing a seamless service. Developing ideas to tackle these challenges requires efficient algorithms that mitigate these issues mentioned above and adapt according to changes in a network environment.

### C. Load Balancing in Hybrid RF/LiFi Systems

Load balancing is essential to ensure a balanced use of the available network resources, particularly in dynamic traffic scenarios. Optimized load balancing techniques improve the overall system efficiency, increase user satisfaction, and reduce congestion [20].

Load-balancing methods often depend on static algorithms that are not adaptable to changing network circumstances.

Many load balancing approaches for hybrid RF/LiFi systems have been discussed thoroughly by the author [21]. Throughput and user satisfaction increase significantly with algorithms that utilize real-time traffic monitoring and responsive resource allocation. The authors demonstrated that round-robin and least connections algorithms efficiently allocated traffic between access points [22], [23]. However, these solutions tend to prove insufficient in dynamic scenarios characterized by time-varying user needs.

In recent years, new approaches have gained traction that encompass machine learning and optimization-based state-of-the-art solutions to load-balancing scenarios. A load-balancing architecture is proposed using deep reinforcement learning for resource allocation in response to the instantaneous network state. The performance showed increased stability and congestion reduction, thus proving that DRL is effective for load-balancing problems in IoT with many dynamics [24], [25]. In addition, many studies investigated the load-balancing-based integration of RF and LiFi technologies. A hybrid load balancing scheme is proposed leveraging both RF and LiFi features to improve the efficiency of the allocation of the resource [26], [27], [28], [29]. Thus, their results showed that these approaches increase both data rates and user capacity, which complement each other in an overcrowded IoT context [30], [31]. Furthermore, many studies used traditional SAC, TRPO, TD3, and A2C algorithms. Resource-constrained devices may struggle with SAC, TRPO, TD3, and A2C in IoT contexts due to their high processing requirements and hyper-parameter tuning sensitivity. Effective training requires steady data streams, which volatile IoT situations often lack. In situations with few rewards, the exploration-exploitation trade-off causes inefficient learning, limiting real-time decision-making. These studies discussed large-scale networks of interconnected devices due to scalability and communication costs [32], [33].

#### D. Challenges in Load Balancing

Although load-balancing strategies have become increasingly sophisticated, some problems have disappeared. The most important of these is access to network status updates, which is not easy on the large scale of IoT. However, the combination of RF and LiFi generates additional complexities in allocating capacity sensibly during contention. Existing approaches are unsuited to the dynamic change in the IoT and require a suitable approach. Moreover, we have also summarized the baseline works in Table I.

### III. SYSTEM MODEL

This study focuses on an indoor IoT scenario, such as a large office or smart building, where hybrid RF/LiFi systems are deployed to support dense device connectivity. The 500 m  $\times$  500 m simulation area Table III represents a typical large indoor environment with multiple access points (APs) strategically placed to ensure coverage and minimize interference. LiFi's high-speed, line-of-sight communication complements RF's broader coverage, addressing the challenges of congestion and variable user demands in such settings. By optimizing user association and load balancing in this indoor context,

TABLE II  
LIST OF KEY NOTATIONS

| Notation        | Definition                                                                           |
|-----------------|--------------------------------------------------------------------------------------|
| $M$             | Markov Decision Process (MDP)                                                        |
| $S$             | State space at time $t$                                                              |
| $A$             | Action space                                                                         |
| $P$             | Transition probability function $P(s_{t+1} s_t, a_t)$                                |
| $R$             | Reward function in a given state                                                     |
| $\gamma$        | Discount factor ( $0 \leq \gamma < 1$ )                                              |
| $s_t$           | State at time $t$                                                                    |
| $a_t$           | Action taken at time $t$                                                             |
| $r_t$           | Reward received at time $t$                                                          |
| $w_1, w_2, w_3$ | Weights for QoS,                                                                     |
| $Q$             | Quality metric vector for each device                                                |
| $L$             | Load on access points                                                                |
| $A_t$           | Indicating which devices association access points                                   |
| $M_t$           | Mobility state of the devices                                                        |
| $U$             | Maximum user capacity of an access point                                             |
| $L_j$           | Load on access point $j$                                                             |
| $T$             | Total throughput in the network                                                      |
| $R_{LiFi}$      | Data rate for the LiFi connection                                                    |
| $SINR$          | Signal-to-Interference-plus-Noise Ratio                                              |
| $I$             | Light intensity received from a LiFi access point                                    |
| $G$             | Channel increase for LiFi                                                            |
| $P_{tx}$        | Transmitted optical power from a LiFi access point                                   |
| $d$             | Distance from the LiFi access point                                                  |
| $P_{rf}$        | Transmit power levels of RF access points                                            |
| $\alpha, \beta$ | Weighting factors in the reward function to balance data rates and power consumption |

our proposed DJHS framework leverages the strengths of both technologies to achieve high throughput, energy efficiency, and scalability.

To analyze and optimize user association and load balancing in a hybrid RF/LiFi IoT environment shown in Fig. 1, we define a comprehensive system model that includes the RF system model, LiFi model, channel model, motion model, communication model, and the problem formulation. The key notations of this work are discussed in Table II. Each component of the system model is described mathematically as follows:

#### A. RF System Model

The RF system is characterized by its coverage area, signal strength, and user capacity, defined as follows:

1) *Coverage Area (C)*: The coverage area of RF access points (APs) is given as

$$C = \{(x, y) \mid d(x, y) \leq R\}, \quad (1)$$

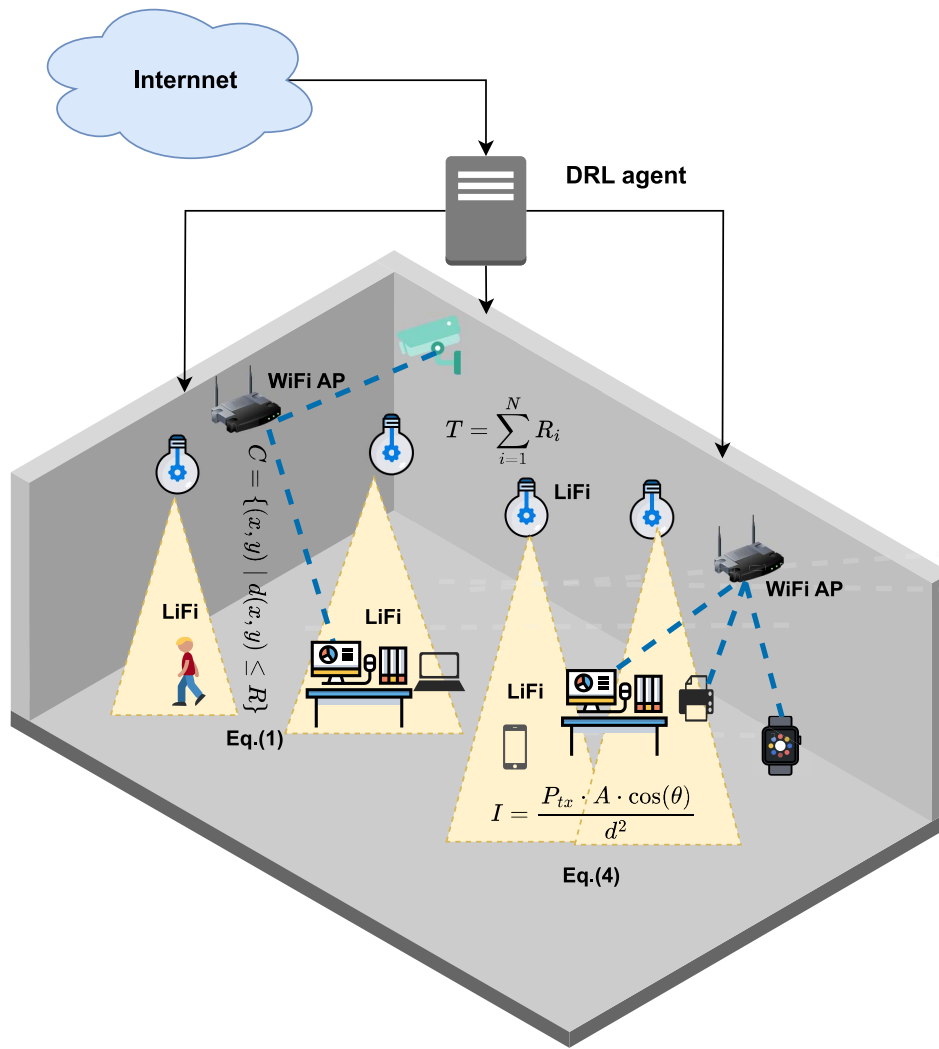
where  $R$  is the RF AP's maximum range and  $d(x, y)$  is the distance between the RF AP and the point  $(x, y)$  in the coverage region.

2) *Signal Power ( $P_r(d)$ )*: The signal strength at a distance  $d$  from the RF access point is represented by the path loss model as

$$P_r(d) = P_0 - 10n \log_{10}(d) - L_w + X_\sigma, \quad (2)$$

where  $P_r(d)$  is the received signal power (in dBm) at a distance  $d$ ,  $P_0$  is the reference signal power at 1 meter (in dBm),  $n$  is the path loss exponent,  $L_w$  is the wall attenuation factor (in dB) accounting for indoor obstacles, and  $X_\sigma \sim$





$N(0, \sigma^2)$  is a Gaussian random variable (in dB) representing log-normal shadowing due to environmental variations. The term  $L_w$  is set based on the number and type of walls between the transmitter and receiver, typically ranging from 5–15 dB per wall, as per indoor propagation models. In Equation (2), the log-distance path loss model uses a path loss exponent  $n \in [2, 4]$ , wall loss  $L_w \in [5, 15]$  dB per wall, and shadowing variance  $\sigma$  derived from standard indoor propagation models, validated against empirical data for large office environments. These parameters are implemented in the 500 m  $\times$  500 m simulation area described in Table III, ensuring realistic modeling of RF channel conditions.

$$U = \frac{E_u}{B \log_2(1 + \text{SINR})}, \quad (3)$$

allocation consistent with the power optimization objectives of the DJHS framework.

The LiFi model describes the characteristics of LiFi communication, focusing on the light source and channel conditions.

$$I = \frac{P_{tx} \cdot A \cdot \cos(\theta)}{d^2}, \quad (4)$$

2) *Channel Gain (G)*: The channel gain ( $G$ ) for LiFi is represented as

$$G = \frac{A \cdot h}{d^2}, \quad (5)$$

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3) *Data Rate*: The attainable data rate for LiFi is expressed as

$$R_{LiFi} = B \log_2(1 + \text{SNIR}), \quad (6)$$

where SNIR is the Signal-to-Noise Ratio for the LiFi connection. Moreover, the data rate for RF is obtained by [34], which is expressed as

$$C = \frac{1}{2} B \log_2 \left( 1 + w \frac{\rho^2 P_t^2}{\sigma^2} \right), \quad (7)$$

where  $w = \frac{e}{2\pi}$ ,  $\rho$  is the responsivity,  $B$  denotes the modulation bandwidth, and  $\sigma^2$  the Gaussian distribution noise power.

### C. Channel Model

Both RF and LiFi channels are included in the channel model, which captures their distinct traits and behaviours. In addition, Equations (2), (4), and (5) assume idealized SINR and power level knowledge for modeling simplicity. In practice, the PPO-based DJHS framework accounts for potential delays and inaccuracies by leveraging real-time state updates, ensuring robust performance under dynamic conditions.

1) *RF Channel Model*: The RF channel is represented by a fading model, such as Rayleigh fading:

$$h_{RFI} \sim \mathcal{CN}(0, \sigma^2), \quad (8)$$

where  $h_{RFI}$  is the total channel gain and  $\sigma^2$  is the variance of the fading.

### D. Mobility Model

This work focuses on the 2-dimensional (2D) environment, and we adopted the Random Waypoint Mobility Model (RWMM). In this model, each IoT device moves within the 500 m  $\times$  500 m simulation area, representing a typical large indoor environment such as an office or smart building. In addition, the RWMM is defined as follows: Waypoint Selection: Each device randomly selects a destination waypoint ( $x, y$ ) within the simulation area, where ( $x, y \in [0, 100]$ ) meters, drawn from a uniform distribution.

1) *Speed*: The device moves toward the selected waypoint at a speed ( $v$ ), which is uniformly distributed in the range ( $[v_{\min}, v_{\max}]$ ), where ( $v_{\min} = 0.5, \text{m/s}$ ) and ( $v_{\max} = 2, \text{m/s}$ ). These values reflect typical pedestrian or device movement speeds in indoor settings.

2) *Pause Time*: Upon reaching the waypoint, the device pauses for a duration ( $t_{\text{pause}}$ ), uniformly distributed in ( $[0, t_{\text{pause, max}}]$ ), where ( $t_{\text{pause, max}} = 10, \text{seconds}$ ). This pause simulates realistic scenarios where devices temporarily remain stationary, such as a device or a sensor collecting data.

3) *Movement in 2D Space*: Given the indoor context, we model device movement in a 2D plane, as the vertical dimension (height) is typically less significant in such environments, such as devices operate at a roughly constant height of 1–2 meters. The position of a device at time ( $t$ ), denoted ( $P_t = (x_t, y_t)$ ), is updated based on its velocity vector ( $v_t$ ) and time increment ( $\Delta t$ ):

$$[P_t = P_{t-1} + v_t \cdot \Delta t], \quad (9)$$

TABLE III  
SIMULATION PARAMETERS

| Parameter                                               | Value                                     |
|---------------------------------------------------------|-------------------------------------------|
| Area Dimensions                                         | 500m $\times$ 500m                        |
| Number of RF APs                                        | 2                                         |
| Number of LiFi APs                                      | 16                                        |
| Random Orientation                                      | Dynamic                                   |
| User Mobility Range                                     | 0 – 1.5 m/s                               |
| Photodetector Area                                      | 1 cm <sup>2</sup>                         |
| LiFi AP Optical Power                                   | 1000 mW                                   |
| RF AP Transmit Power                                    | 200 mW                                    |
| RF Channel Bandwidth                                    | 20 MHz                                    |
| LiFi Channel Bandwidth                                  | 40 MHz                                    |
| Noise Figure (RF)                                       | 10 dB                                     |
| Noise Figure (LiFi)                                     | 14 dB                                     |
| RF Noise Power Cal                                      | Based on BW and temp                      |
| LiFi Noise Power Cal                                    | Based on BW, temp, and light              |
| Coverage Area Cal Method                                | Based on power and env                    |
| Semi-Angle of LiFi APs                                  | 60°                                       |
| Field of View of Users' Photodetectors                  | 90°                                       |
| Simulation Algorithm Library                            | Stable Baselines3                         |
| No. of Episode                                          | 1000                                      |
| Learning Rate                                           | 0.0007                                    |
| Discount Factor                                         | 0.9                                       |
| AP Height                                               | 5 m                                       |
| <b>Random Waypoint Mobility Model (RWMM) Parameters</b> |                                           |
| Speed Range                                             | ([0.5, 2], m/s)                           |
| SPause Time                                             | ([0, 10], seconds)                        |
| Point Distribution                                      | Uniform over ([0, 100] $\times$ [0, 100]) |
| Boundary Handling                                       | Uniform over ([0, 100] $\times$ [0, 100]) |

where ( $v_t$ ) is computed based on the direction to the current waypoint and the selected speed. After reaching a waypoint and pausing, the device selects a new waypoint, and the process repeats. To ensure realistic mobility patterns, we impose boundary constraints: if a device's trajectory intersects the simulation area's boundaries ( $x = 0, x = 500, y = 0, y = 500$ ), it is reflected back into the area, simulating walls or physical barriers in the indoor environment. Additionally, we incorporate a minimum distance between consecutive waypoints to avoid overly short movements that may not reflect practical device behavior. The RWMM parameters are discussed in Table III. This model ensures that device mobility is both realistic, incorporating indoor mobility with boundary reflections, computationally tractable and accounts for obstacles, while its stochastic nature captures varied device movements. Although clustering is not explicitly modeled, DJHS's real-time adaptability ensures robust performance across mobility patterns.

### E. Communication Model

The communication model describes the devices communicating with either RF or LiFi APs.

1) *Throughput (T)*: The total throughput is represented as

$$T = \sum_{i=1}^N R_i \quad (10)$$

where  $R_i$  denotes the data rate of the  $i^{\text{th}}$  user, and  $N$  represents the total number of users.

2) *User Association*: The association of user  $i$  with access point  $j$  is denoted as a binary variable  $x_{ij}$ :

$$x_{ij} = \begin{cases} 1 & \text{if user } i \text{ is linked to AP } j \\ 0 & \text{otherwise} \end{cases} \quad (11)$$

#### F. Problem Formulation

We aim to improve data throughput and minimize load balancing. The load balancing and user association model optimizes the association of users to RF and LiFi access points.

1) *Load on Access Point ( $L_j$ )*: The load on the access point  $j$  is defined as

$$L_j = \sum_{i=1}^N x_{ij} \cdot \text{Tr}_i, \quad (12)$$

where  $\text{Tr}_i$  represents the traffic demand of user  $i$ .

2) *Load Balancing Objective*: Our objective is to reduce load imbalance among the access points as

$$\min J = \sum_{j=1}^M (L_j - \bar{L})^2, \quad (13)$$

where  $(M)$  is the number of access points and  $(\bar{L})$  be the average load of all access points.

3) *Formulated Association Optimization*: The optimization problem is as follows:

$$\max \sum_{i=1}^N \sum_{j=1}^M x_{ij} \cdot R_{ij}, \quad (14)$$

$$\text{Subject to: } \sum_{j=1}^M x_{ij} \leq 1 \quad \forall i, \quad (15)$$

$$\sum_{i=1}^N x_{ij} \leq U_j \quad \forall j, \quad (16)$$

where  $R_{ij}$  is the achievable data rate for user  $i$  associated with access point  $j$ , and  $U_j$  is the maximum user capacity for access point  $j$ . In addition, handover penalties and queuing dynamics are implicitly addressed by the proposed DJHS framework's real-time state updates, which adapt to user mobility and traffic demands.

#### IV. PROPOSED FRAMEWORK FOR USER ASSOCIATION AND LOAD BALANCING

We propose a novel framework to address the complicated user association and load balancing issues in hybrid RF/LiFi systems, as shown in Fig. 2. In this framework, we optimized network performance, data throughput, and energy efficiency by dynamically adjusting user associations and resource allocations. Our DJHS model, specifically built for continuous action spaces and efficient decision-making settings, enables this real-time adaptability to changing network conditions. We also designed two Algorithms. Algorithm 1 is designed for power allocation, and Algorithm 2 is designed for user association and load balancing.

#### A. Proposed DJHS Method

We employ a DRL-based approach effective in tackling complex action spaces with policy learning. The DJHS model includes the following elements:

*State Space ( $S$ )*: The state space ( $S$ ) maintains all dynamic information of the network and different conditions of the user that are essential for making decisions. ( $S$ ) is given as follows:

$$S = \{(P_{\text{RF},i}, P_{\text{LiFi},i}, x_j, C_{ij}, D_j, L_j, Q_j, \lambda_j, B_j) \mid i \in \mathcal{N} \cup \mathcal{M}, j \in \mathcal{U}\} \quad (17)$$

where  $P_{\text{RF},i}$ ,  $P_{\text{LiFi},i}$  are transmit powers,  $x_j$  is user position,  $C_{ij}$  is channel state,  $D_j$  is delay,  $L_j$  is AP load,  $Q_j$  is queue length,  $\lambda_j$  is task arrival rate, and  $B_j$  is delay buffer occupancy. Incorporating  $Q_j$ ,  $\lambda_j$ , and  $B_j$  enables delay-aware decisions, enhancing QoS in IoT settings, consistent with real-time updates in the proposed algorithm.

*Action Space ( $A$ )*: The action space is continuous, allowing soft association and power control:

$$A = \{(x_{ji}, P_i) \mid x_{ji} \in [0, 1], \sum_{i=1}^M x_{ji} = 1, P_i \in [0, P_{\max}], \forall j \in \text{Users}, i \in \text{APs}\}, \quad (18)$$

where  $x_{ji}$  is the association probability for user  $j$  to AP  $i$ , and  $P_i$  enables joint power control, with LiFi power adjustments approximating beam steering to optimize SINR.

*Reward Function*: The reward function balances throughput, power efficiency, and SLA constraints:

$$R(s, a) = \alpha \left( \sum_{j \in \mathcal{U}} R_{ij} \right) - \beta \left( \sum_{i \in \mathcal{N} \cup \mathcal{M}} P_i \right) - \gamma \sum_{j \in \mathcal{U}} \max(0, D_j - T_j) \quad (19)$$

where  $\alpha$ ,  $\beta$ ,  $\gamma$  are weights,  $R_{ij}$  is data rate,  $P_i$  is power,  $D_j$  is delay, and  $T_j$  is the SLA threshold.

$$S = \{(P_{\text{RF},i}, P_{\text{LiFi},i}, x_j, C_{ij}, D_j, L_j) \mid i \in \mathcal{N} \cup \mathcal{M}, j \in \mathcal{U}\}, \quad (20)$$

where  $P_{\text{RF},i}$  and  $P_{\text{LiFi},i}$  indicates the existing transmit power levels of LiFi and RF access points (APs), respectively.  $x_j$  represents the user's Coordination  $j$  within the network.  $C_{ij}$  is current channel state information between AP  $i$  and user  $j$ , with signal strength and quality metrics. Active user  $j$  for required data rate  $D_j$ .  $L_j$  is AP load  $i$ , representing the total traffic demands. In this regard, we design a DJHS framework for user association and load balancing in hybrid RF/ LiFi systems that overcome the problems caused by the network conditions and different user requirements. The overarching goal of this framework is to maximize network throughput while maintaining energy efficiency by appropriately utilizing  $S$  along with a carefully designed  $R$ . The reward function in Algorithm 1 integrates throughput, power efficiency, and QoS, implicitly promoting fairness by minimizing load imbalance across APs, and dynamically adapts to network conditions by leveraging real-time state variables such as user positions, channel states, and AP loads, ensuring optimal performance in dynamic IoT environments. To ensure stable and efficient policy updates in the DJHHS framework, our PPO

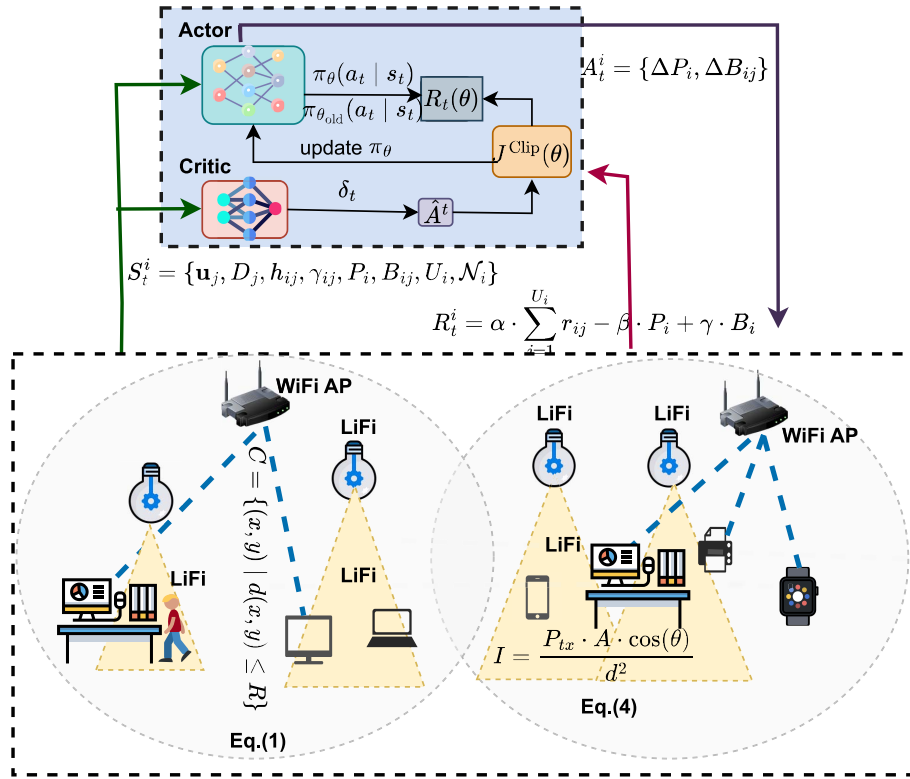


Fig. 2. Proposed Framework for Hybrid RF/LiFi Environment in IoT.

implementation incorporates value clipping and generalized advantage estimation (GAE). Value clipping, with a clipping parameter  $\epsilon = 0.2$ , constrains the value function updates to prevent large deviations in the policy, enhancing training stability in the dynamic IoT environment. GAE, with a discount factor  $\gamma = 0.99$  and lambda  $\lambda = 0.95$ , is used to estimate advantages, balancing bias and variance to provide robust advantage estimates for policy optimization. These components, implemented via Stable Baselines3, contribute to the consistent convergence observed across varying user densities and SINR levels.

### B. DJHS Algorithm Overview

To solve the problems mentioned in Equations (13)-(15), we designed a DRL based DJHS algorithms. This study aims to improve the policy by taking the largest possible improvement step while avoiding destabilizing the learning process. This is achieved through a clipped objective function that penalizes large changes to the policy. We consider that in PPO, we updated our policy based on new data. We define a probability ratio as

$$r(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{old}}(a_t | s_t)} \quad (21)$$

where  $\pi_\theta(a_t | s_t)$  is the probability of taking action  $a_t$  in state  $s_t$  under the *new* policy (parameterized by  $\theta$ ). The  $\pi_{\theta_{old}}(a_t | s_t)$  is the probability of taking action  $a_t$  in state  $s_t$  under the *old* policy (parameterized by  $\theta_{old}$ ). The probability ratio  $r(\theta)$  essentially tells how much the new policy's probability of taking a particular action differs from the old policy's

probability. A value of  $r(\theta) = 1$  means the new policy agrees with the old policy on that action. A value greater than 1 means the new policy is more likely to take that action, and a value less than 1 means it's less likely. The clipped objective function in PPO then limits how far this ratio deviates from 1. Specifically, it clips the ratio to be within a certain range, such as  $[1 - \epsilon, 1 + \epsilon]$ , where  $\epsilon$  is a hyperparameter). This clipping prevents the policy from changing too drastically in a single update, which can lead to instability. The policy that moves the probability ratio away from 1" means that the PPO algorithm reduces the reward when the new policy suggests actions that are very different from what the old policy would have done. The policy that moves the probability ratio away from 1" means that the PPO-based algorithm reduces the reward when the new policy suggests actions that are very different from what the old policy would have done. These are more conservative and stable updates. The reward function of the proposed Algorithm 1 effectively balances throughput and power efficiency through a weighted multi-objective reward, enabling expressive policy learning. Moreover, this supports service-level agreement (SLA) requirements through QoS optimization and load balancing, ensuring reliable connectivity and low latency for diverse IoT applications.

### C. Complexity Analysis of the Proposed DJHS Algorithms

The complexity of the proposed DJHS framework for user association and load balancing in hybrid RF/LiFi networks is described by analyzing the  $S$ ,  $A$ ,  $R$ , transition probability, and the proposed DJHS based on PPO algorithm computations. The state space  $S$  encompasses data on user-AP associations,



**Algorithm 1** DJHS-Based Power Allocation Using PPO in Hybrid RF/LiFi Systems

- 1: Initialize the network with sets of LiFi Access Points (APs)  $\mathcal{N}$ , RF APs  $\mathcal{M}$ , and user equipment  $\mathcal{U}$ .
- 2: Specify the state space  $\mathcal{S}$  to contain data needs, AP power levels, user positions, and signal to interference plus noise ratio (SINR).
- 3: Define the action space  $\mathcal{A}$  for continuous adjustment of transmit power levels  $P_i^{rf}$  for RF APs and  $P_i^{Lifi}$  for LiFi APs.
- 4: Define the reward function  $R(s, a)$  to balance network throughput and power efficiency, considering energy consumption and user QoS.
- 5: **while** training episodes  $\leq$  total episodes **do**
- 6:   Reset the environment to the initial state  $s_0$ .
- 7:   **for** each step  $t$  in the episode **do**
- 8:     Observe the current state  $s_t$  from the environment.
- 9:     Select action  $a_t$  using the PPO policy  $\pi_\theta(a_t|s_t)$ , where  $a_t$  adjusts the power levels of APs.
- 10:    Apply action  $a_t$ : adjust AP power levels accordingly for both RF and LiFi APs.
- 11:    Receive new state  $s_{t+1}$  and the corresponding reward  $r_t$  based on network throughput and energy usage.
- 12:    Update the PPO model by optimizing the clipped surrogate objective function to improve policy  $\pi_\theta$ .
- 13:   **end for**
- 14:   Adjust the PPO policy  $\pi_{PPO}$  after each episode based on the policy updates.
- 15: **end while**

load metrics, and signal quality, with a complexity as

$$O(S) = O(N \cdot M \cdot d), \quad (22)$$

where  $N$  is the number of users,  $M$  the number of access points, and  $d$  the dimensionality of state information, including queue length, task arrival rate, and delay buffer occupancy for delay-aware decisions. The action space, responsible for managing user associations and load distribution, scales with as

$$O(A) = O(N \cdot M), \quad (23)$$

The above equation presents a high-dimensional space for the DRL agent to navigate for scaling. Additionally, each transition in the network state requires real-time adjustments based on user movements and interference, with transition probability complexity approximated as

$$O(P) = O(N \cdot M^2), \quad (24)$$

Moreover, the reward function's complexity,  $O(R) = O(N \cdot M)$ , integrates multiple objectives such Quality of Service (QoS), load balance, and energy consumption. This reward function guides the agent in optimizing both user experience and resource efficiency. Leveraging DJHS to handle these high-dimensional states and actions, the framework's total complexity is again as

$$O(\text{Total}) = O(E \cdot S \cdot A \cdot \text{NN}_{\text{pm}}), \quad (25)$$

**Algorithm 2** Dynamic User Association and Load Balancing in Hybrid RF/LiFi Networks

- 1: **procedure** MANAGE-NETWORK( $\mathcal{U}, \mathcal{B}, \mathcal{C}$ )  $\triangleright$  Users  $\mathcal{U}$ , Access Points  $\mathcal{B}$ , Channels  $\mathcal{C}$
- 2:   **for all**  $u \in \mathcal{U}$  **do**  $\triangleright$  Iterate over all users
- 3:     UpdatePosition( $u$ )  $\triangleright$  Update the position of the user  $u$  based on mobility patterns
- 4:      $c^* \leftarrow$  Assign-Channel( $u, \mathcal{C}$ )  $\triangleright$  Assign the best channel  $c^*$  to user  $u$  based on interference and availability of channels  $\mathcal{C}$
- 5:     Update-SINR( $u, c^*$ )  $\triangleright$  Calculate the SINR for user  $u$  on the assigned channel  $c^*$
- 6:     Adjust-Power( $u, c^*$ )  $\triangleright$  Adjust the transmission power for user  $u$  on channel  $c^*$  to optimize energy usage and throughput
- 7:   **end for**
- 8:   Update-Network-Parameters()  $\triangleright$  Update global network parameters after processing all users
- 9: **end procedure**
- 10: **function** ASSIGN-CHANNEL( $u, \mathcal{C}$ )  $\triangleright$  Assign the most suitable channel to the user  $u$
- 11:   **return** Channel  $c$  with minimum interference and within the coverage area of user  $u$
- 12: **end function**
- 13: **function** UPDATE-SINR( $u, c$ )  $\triangleright$  Calculate the SINR for the user  $u$  on channel  $c$
- 14:   Calculate SINR using the received power, interference from neighboring users, and noise on channel  $c$
- 15: **end function**
- 16: **function** ADJUST-POWER( $u, c$ )  $\triangleright$  Adjust the transmission power based on the calculated SINR
- 17:   Adjust the transmission power of user  $u$  on channel  $c$  to balance energy consumption and achieve the required data rate
- 18: **end function**

where  $E$  is the number of episodes, and  $\text{NN}_{\text{pm}}$  represents the number of neural network parameters used in PPO-based DJHS, includes the entropy term ( $\beta = 0.01$ ) for exploration, with negligible computational overhead, ensuring stability in dynamic topologies. This complexity is substantial due to the demands of continuous policy adjustments and neural network back-propagation, especially in large-scale IoT environments where network load and mobility patterns fluctuate considerably. The stability of this high-dimensional computation is ensured by PPO's clipped surrogate objective, augmented by value clipping ( $\epsilon = 0.2$ ) and GAE ( $\gamma = 0.99$ ,  $\lambda = 0.95$ ), which constrain policy updates and provide robust advantage estimates, respectively, enabling effective navigation of the vast state-action space in dynamic IoT environments.

Overall, the framework's complexity reflects the intricate balancing act of real-time resource allocation in hybrid RF/LiFi networks, where the proposed method of DJHS adaptive learning enables it to handle the vast state-action space and maintain optimal network performance under variable conditions.

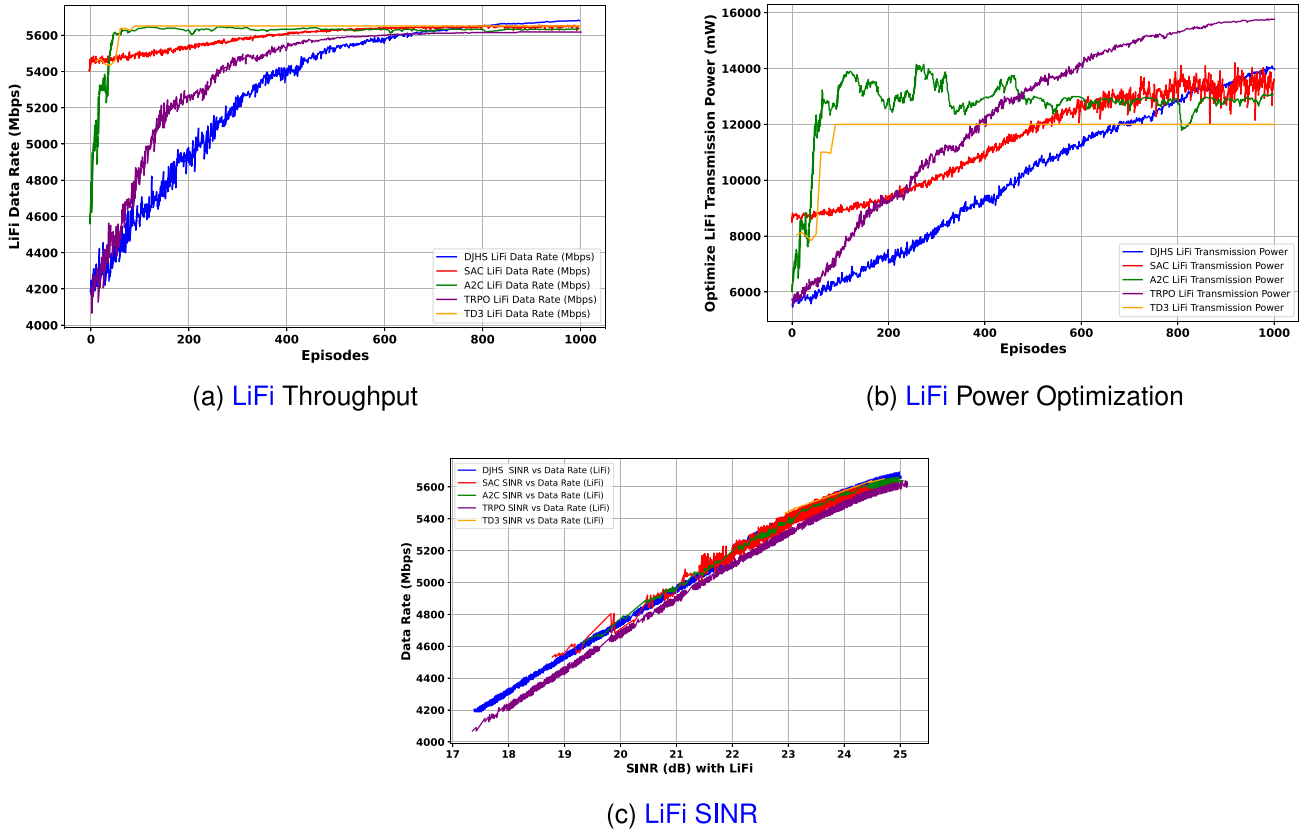


Fig. 3. Comparison of Proposed DJHS with baseline algorithms for throughput, power optimization and SINR concerning LiFi.

## V. PERFORMANCE EVALUATION

This section presents a comprehensive performance evaluation of our proposed DJHS framework, which is designed to improve load balancing and user association and optimize power allocation performance in a hybrid RF/LiFi communication system. The analysis is conducted using traditional reinforcement learning algorithms such as A2C, TRPO, TD3 and SAC [35]. These methods are benchmarks against our proposed DJHS method. Moreover, the sensitivity analysis is performed by varying user density (10–50 users), AP placements (5–10 APs), and obstacle effects via RWM boundary reflections in the 500 m  $\times$  500 m simulation area. The results confirm DJHS's consistent throughput and energy efficiency across these conditions, supporting its generalizability.

### A. Simulation Scenario

The purpose of this simulation is to implement a hybrid wireless fidelity (WiFi) and LiFi network environment shown in 2, where it presents the user association and load balancing methods. The environment covers a typical 500 m  $\times$  500 m area for IoT scenario. We deploy APs mentioned in Table III, 900-3600 channels, and 10-100 users. The users are spawned in a random location, with randomized velocity within those bounds. Random velocities (x and y) so movement can occur in different directions. Strategically locating such APs is critical to achieving a blanket coverage and continuous connection effect. WiFi APs are centrally located to provide optimal coverage of RF signals, while LiFi access points

have been distributed across the room to prevent overlapping and interference between light-based communications streams. We model not just with a realistic IoT environment but also consider various user mobility and coverage characteristics. Moreover, we used a Python-based simulation framework developed to predict the behavior of hybrid RF/LiFi networks accurately. A complete description of this simulation setup is discussed in Table III.

### B. Performance Comparison of LiFi With Episodes

This section discusses the effectiveness of our proposed algorithm, DJHS, compared to other baseline algorithms. The results show that our proposed DJHS outperforms all baseline algorithms regarding LiFi. The results are discussed in Fig. 3.

1) *LiFi Data Throughput*: Fig. 3a illustrates the data rates achieved by various reinforcement learning methods in a LiFi communication context, with a noticeable performance advantage observed in the DJHS method. DJHS outperforms the baseline methods, SAC, A2C, TD3, and TRPO, throughout the episodes, indicating its superior efficiency in optimizing data transmission rates. DJHS's ability is relatively more robust owing to its adaptive learning strategies and remarkable exploration functions that lead to efficient navigation of the complexities of LiFi. Local optima may challenge the other used methods for a lengthy convergence, as DJHS effectively balances exploration and exploitation, offering better performance. Therefore, this yields a stronger data rate that also demonstrates the DJHS capability as a potential solution

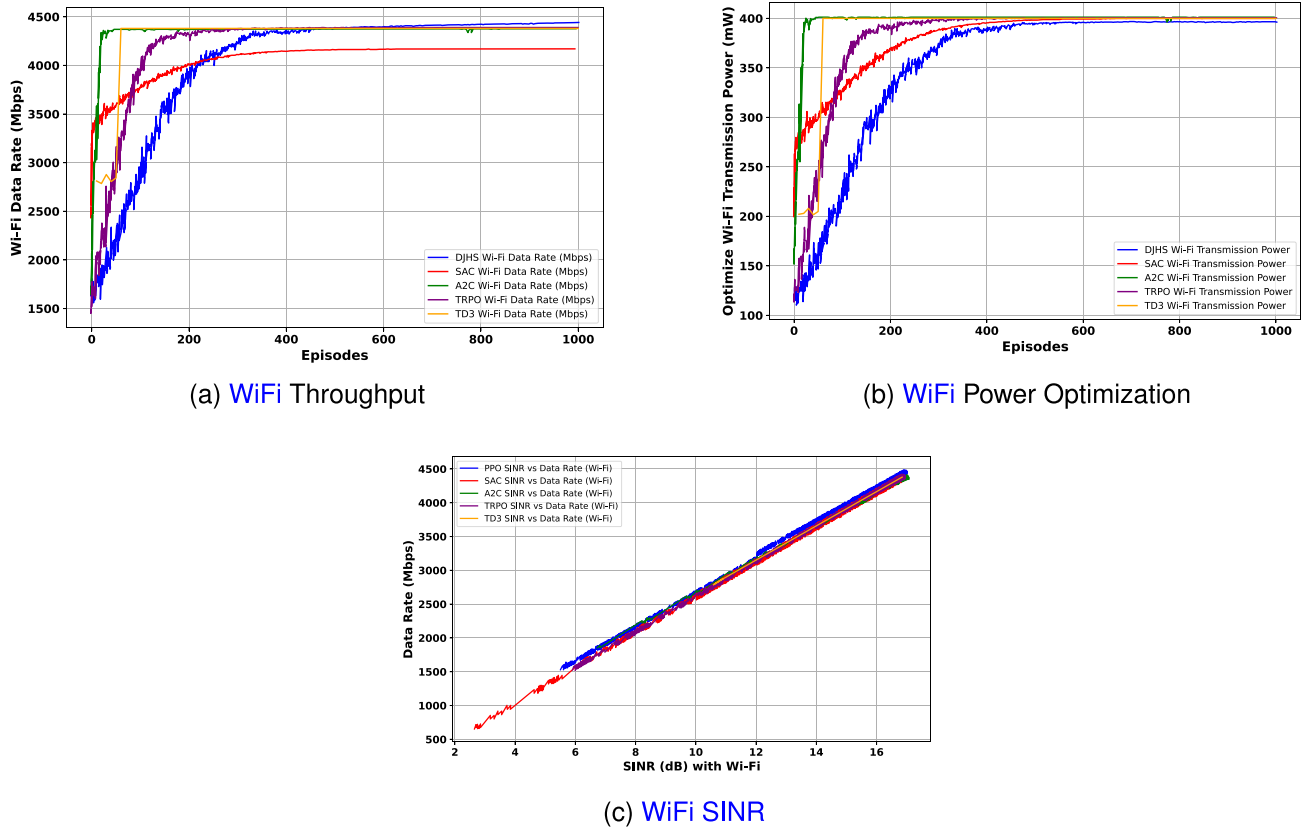


Fig. 4. Comparison of Proposed DJHS with baseline algorithms for throughput, power optimization and SINR concerning WiFi.

for real-world LiFi applications in which high throughput and dependability are of key importance.

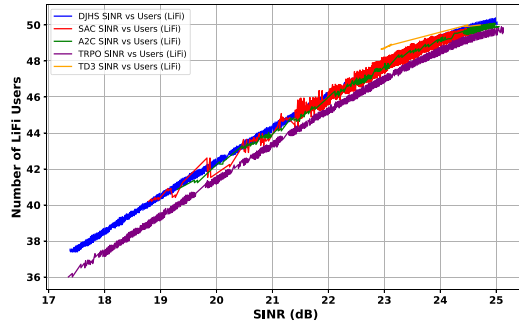
2) *Power Optimization*: The findings in Fig. 3b underscore the optimization of LiFi transmission power across multiple episodes, with the DJHS approach exhibiting a notable performance superiority over the baseline methods, specifically SAC, A2C, TD3 and TRPO. DJHS consistently attains reduced transmission power, demonstrating its efficacy in enhancing power efficiency while ensuring dependable communication. This benefit is ascribed to DJHS's sophisticated power management strategy, which integrates real-time feedback and adaptive learning techniques to reduce energy consumption while maintaining data quality. Conversely, alternative solutions generally exhibit diminished adaptability and sometimes depend on static strategies that inadequately optimize power consumption. Consequently, DJHS not only diminishes operational expenses related to energy consumption but also improves the sustainability of LiFi systems, rendering it an optimal selection for energy-efficient data transmission in practical applications.

3) *LiFi SINR*: Figure 3c depicts the correlation between SINR (Signal to Interference and Noise Ratio) and data rates attained by various methods, with the DJHS method noticeably surpassing the baseline techniques of SAC, A2C, TD3, and TRPO. DJHS demonstrates enhanced data rates across different SINR levels, highlighting its resilience in sustaining high performance in suboptimal situations. This advantage arises from DJHS's capacity to balance policy updates adeptly using truncated surrogate functions, stabilizing

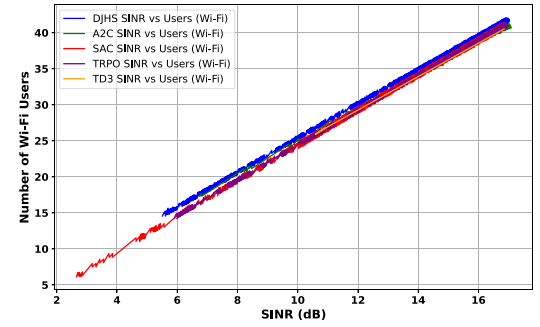
training, and improving convergence to optimal policies. On the other hand, the SAC, A2C, TD3, and TRPO, which face challenges such as instability in policy updates or slower response to changes in SINR, DJHS adapts more fluidly to varying channel conditions. This adaptability leads to higher data rates and ensures reliable communication in environments with fluctuating interference, making DJHS the most effective method for optimizing performance in LiFi systems under the given conditions.

### C. Performance Comparison of WiFi With Episodes

1) *WiFi Data Throughput*: Fig. 4a displays a direct comparison of the WiFi data rates over episodes for different reinforcement learning methods (SAC, A2C, TD3, and TRPO) against the new reinforcement learning methods proposed in this study (DJHS), and shows that DJHS outperforms all three baseline methods consistently. Throughout the episodes, DJHS consistently shows significantly higher data rates, indicating its effectiveness in utilizing WiFi. Such a great superiority is due to the fact that DJHS is utilizing exploration that is more evolved, and learning algorithms are also slightly sophisticated, enabling it to highly adapt and tune its actions based on real data feedback in time. However, the baseline methods are relatively slower to converge and get trapped in local optimums, causing poor transmission performance. DJHS, which combines dynamic policy change along with an accelerated learning approach, not only improves the maximum data throughput but also increases the average stability of a WiFi communicational link, making it most



(a) LiFi SINR with Users



(b) LiFi SINR with Users

Fig. 5. Comparison of Proposed DJHS with baseline algorithms concerning users.

popular for practical applications where higher rates need to be achieved.

2) *WiFi Power Optimization*: The power optimization results are shown in Fig. 4b. We notice that DJHS outperforms all of the baseline methods SAC, A2C, and TRPO efficiently across different episodes for optimizing WiFi transmission power. This shows that DJHS generates the lowest values of transmission power; therefore, it is useful for having lower values of total power consumption while maintaining a high level of reliability in communication. DJHS achieves this outperformance due to its adaptive learning capabilities, which means it continuously modifies its strategy depending on the information received from the environment. In contrast to the numerous other approaches that have traditionally depended on stiffer or less adaptive techniques, DJHS strikes a balance between exploration and exploitation such that it quickly finds and holds power settings that are better than any previously found. The reduction in the WiFi energy cost provided by DJHS then turns it into a more sustainable and efficient means of communication, hence making it a practical device for applications whose objective is power improvement.

3) *WiFi SINR*: Fig. 4c shows the relationship between SINR and the data rates achieved by various methods in a WiFi scenario, where the DJHS method outperforms the baseline baselines such as SAC, A2C, TD3, and TRPO by a large margin. DJHS consistently achieves better data rates across the SINR range, demonstrating its effectiveness in improving performance under challenging conditions. The reasons behind this supremacy can be attributed to advanced adaptive learning algorithms based on dynamic policy adaptation through a real-time received SINR, which enables DJHS to leverage beneficial conditions whilst minimizing interference effects. On the other hand, while the alternative methods are functional, they may take a longer time to adapt to changing SINR and, therefore, may provide lower-than-expected data rates. By means of targeted research and accurate policy updates, DJHS effectively utilizes throughput whilst increasing the reliability of WiFi transmissions, making it the best method for achieving higher data rates in real practice.

#### D. Performance Comparison With Number of Users

1) *LiFi SINR*: Fig. 5a shows SINR and user support in a LiFi setting, which demonstrates an overall comparison

of DJHS w.r.t baseline methods, namely, SAC, A2C, TD3 and TRPO. DJHS also serves more users at different SINR levels in each band compared to the other two algorithms, which confirms its potential to manage concurrent connections effectively. DJHS outperforms the other two simulated approaches in part due to its dynamic assignment of resources that dynamically optimizes given feedback from the network environment. DJHS, on the other hand, is efficient with various other methods that struggle when user load increases, and it also has mechanisms to balance resource allocation per connection to make sure all connections get good quality service. Thus, DJHS provides the capability of handling more users while offering enhanced reliability to the wireless network, and it is the favourite method for supporting more users in LiFi systems.

2) *WiFi SINR*: The results of WiFi SINR are shown in Fig. 5b, which indicates a connection between SINR and the number of users that connect to a WiFi network. The DJHS method is clearly better than the baseline methods SAC, A2C, TD3, and TTRPO. DJHS works well at increasing the number of users that connect to WiFi networks because it can handle a lot more users at different SINR levels. This better performance is due to DJHS's advanced adaptive resource management strategies, which let it change how bandwidth is used and how transmission settings are optimized based on what's happening in real time. On the other hand, the different ways might have trouble allocating resources and keeping users from using it too much at once, which could lower the number of users that use the service and make it less reliable. DJHS improves network performance by effectively balancing the needs of multiple users while keeping high data rates. This makes it the best choice for WiFi systems that need to support more users.

## VI. CONCLUSION

The DJHS method represents a significant advancement in addressing the intertwined challenges of user association, power optimization, and load balancing in hybrid RF/LiFi systems for IoT applications. As the number of devices increases, traditional methods struggle to maintain efficiency, often leading to suboptimal data rates and power consumption performance. The empirical results of this study demonstrate that DJHS not only achieves higher data rates and better



power efficiency but also supports a greater number of users compared to conventional reinforcement learning algorithms like SAC, A2C, TD3, and TRPO. The adaptive nature of DJHS allows it to dynamically optimize resource allocation based on real-time network conditions, significantly enhancing overall system performance and user satisfaction. By effectively balancing user demands and minimizing energy consumption, DJHS addresses critical scalability issues in IoT environments. Furthermore, its capability to manage varying SINR conditions ensures reliable connectivity and robust communication, which are vital for the seamless operation of IoT applications. Moreover, the potential for further enhancements in DJHS is promising. Future research can focus on integrating hierarchical and multi-agent reinforcement learning techniques and exploring federated learning for decentralized training. These enhancements will enhance efficiency and scalability, reinforcing DJHS's position as a premier solution for the intricate issues confronting hybrid wireless communication systems. This study highlights the significance of new strategies in enhancing IoT networks, hence facilitating more sustainable and efficient communication technologies.

#### APPENDIX

The PPO algorithm in the DJHS framework leverages value clipping and generalized advantage estimation (GAE) to ensure stable policy updates. Value clipping is applied to the value function loss, defined as

$$L^{\text{CLIP}} = \min\left((V_{\theta}(s) - V_t)^2, \text{clip}(V_{\theta}(s) - V_t, -\epsilon, \epsilon)^2\right),$$

with  $\epsilon = 0.2$ , to limit large updates. GAE is computed as

$$A_t = \sum_{t'=t}^{\infty} (\gamma\lambda)^{t'-t} \delta_{t'},$$

where  $\delta_{t'} = r_{t'} + \gamma V(s_{t'+1}) - V(s_{t'})$ , with  $\gamma = 0.99$  and  $\lambda = 0.95$ . These settings, implemented via Stable Baselines3, ensure robust convergence in our simulations, with hyperparameters including a learning rate of 0.0003, batch size of 64, and two hidden layers of 128 units each.

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